WANG ZHAODONG

Pronounced roughly: Wong Jow-dong

Design Researcher
Master's Candidate in Design

+86 133 2708 9523
tungwang@tongji.edu.cn
Shanghai, China
0009-0008-1111-6742
LinkedIn
cv.tung.mockingbird.team

Design researcher investigating and reimagining social relations in shared environments through discursive design, embodied inquiry and technological mediation.

RESEARCH INTERESTS

Public Space and Social Relations;
Discursive and Speculative Design;
Embodied Experience and Meaning-Making;
Human–Technology Relations and Design Methodology

RESEARCH CAPABILITIES

First-Person / Qualitative Inquiry
Conceptual & Relational Modelling
Empirical-to-Design Translation
Metaphor/Schema Analysis • XR Prototyping

PUBLICATIONS & MANUSCRIPTS

Z. Wang, & F. You (2026). From Subtle Embodied Social Experience to Designable Relations: An Image-Schema-Mediated Empirical-to-Design Method
First author • Under review at *ACM CHI 2027* • Preprint available on SSRN
doi:10.2139/ssrn.7201181

F. You, X. Yan, **Z. Wang**, & Q. Fu (2026). Seeing the whole process: How cooperative process transparency in AR-HMI affects driver cognition in V2V lane changes.
Corresponding author • *Transportation Research Part F: Traffic Psychology and Behaviour*, 123, 103830.
doi:10.1016/j.trf.2026.103830

Y. Chen, F. Jia, **Z. Wang**, J. Zhang, F. You, & J. Wang (2026). Designing multimodal takeover request interfaces: The impact of warning modalities and information types on drivers' takeover performance and subjective perception.
Co-author • Under review at *Transportation Research Part F: Traffic Psychology and Behaviour*

Y. Chen, Y. Jin, F. Jia, J. Wang, **Z. Wang**, B. Mehler, & F. You (2026). Beyond attention reminders: Effects of monitoring request design type and non-driving-related task engagement on driver takeover performance and subjective perceptions.
Co-author • Under review at *Accident Analysis & Prevention*

Y. Chen, J. Wang, F. Jia, X. Wu, Q. Xiao, **Z. Wang**, & F. You (2025). Is control necessary for drivers? Exploring the influence of human–machine collaboration modes on driving behavior and subjective perception under different hazard visibility scenarios.
Co-author • *Accident Analysis & Prevention*, 217, 108067. doi:10.1016/j.aap.2025.108067

EDUCATION

9/2024 - 6/2027 **M.A. in Design** — **Tongji University, Shanghai, China**
Working thesis title: *Mixed-Reality Interaction Design for Public-Space Social Experience*
GPA: 4.76/5.0 • Supervisor: Professor Fang You, FRSA
Selected coursework: User Study 98/100; Narratology and Storytelling 90/100; Service Design 96/100.

9/2020 - 6/2024 **B.Sc. in Physics** — **Jilin University, Changchun, China**
Selected coursework: Computational Physics A 96/100; Mechanics 93/100; C Programming 92/100.

SELECTED RESEARCH PROJECTS

2025–Present **Subtle Embodied Social Experience and Design** — **Master's Thesis Researcher, Tongji University**

- **Social and design proposition:** Investigate subtle embodied social experience as material for discursive design and social imagination, using stranger gaze as a case to reimagine public co-presence; employ public-space XR imagery as an interaction mediator and discursive carrier.
- **Experiential modelling:** Drawing on cognitive linguistics, developed a theory-informed analytical model of first-person meaning construal of perceived gaze exchange to inform symbolic metaphor design; distinguished substrate-level from meaning-construal affordances.
- **First-person inquiry:** Established an episode-focused, micro-phenomenologically informed first-person inquiry into experiential structures, with separate perceptual and metaphor analyses distinguishing problem-defining conditions from relation-providing conceptual-metaphorical relations.
- **Empirical-to-design translation:** Developed an image-schema-mediated empirical-to-design method based on first-person inquiry and cognitive linguistics, making experiential material available for design by structuring it as a constrained relational design problem.
- **Design case and implementation:** Applied the method with 16 participants (782 interview minutes; 160,162 Chinese transcript characters), deriving a relational grammar and three XR propositions; developed Membrane–Bubble in Unity and deployed it on Meta Quest Pro for a planned laboratory study.

2025 - Present **Cooperative Process Transparency in AR-HMI** **Co-Researcher, Tongji University**

Examines how different levels of cooperative process transparency affect driver cognition during Vehicle-to-Vehicle (V2V) cooperative lane changes.

- Contributed to the conceptualization and three-level operationalization of cooperative process transparency; curated and formally analysed performance and subjective data.
- Produced research visualizations and led manuscript preparation and revision.

2024 - 2025 **AR Social-Space Design through Interpersonal Distance and Social Cues** **Co-Researcher, Tongji University**

Examines how virtual-agent layout, gesture amplitude, and interaction guidance shape affect, social presence, and continued engagement in AR.

- Contributed to research conceptualization and curated and formally analysed the experimental data.
- Drafted the original manuscript and contributed to its review and revision.

2024 - Present **Automated-Driving HMI and Driver Attention** **Co-Researcher, Tongji University**

Comprises three driving-simulator studies of collaboration modes, monitoring requests, and multimodal takeover warnings.

- Contributed to the design and execution of one controlled driving-simulator study; processed and analysed eye-tracking and fNIRS data across the project studies.
- Contributed to the interpretation of results and manuscript review and revision.

SELECTED DESIGN & SPATIAL PRACTICE

2023–Present **ExpoCraft – Reconstruction of Expo 2010 Shanghai** **Independent Designer · Ongoing Project**

Reconstruct Expo 2010 Shanghai in Minecraft from maps, photographs and archival records; translate fragmented sources into a navigable spatial archive.

2025 **The Field of Seeds – Physical-Computing Installation** **Co-Designer · Course Project**

Developed the concept, analysed the site and designed interactions/circuitry; co-built a wind/proximity-responsive Arduino prototype inspired by the UK Pavilion (Seed Cathedral) at Expo 2010 Shanghai.

2025 **Rooftop Garden – Site-Specific Interactive Projection** **Independent Designer · Course Proposal**

Designed a projection proposal for a typhoon-displaced roof: passersby's silhouettes trigger projected flowers, linking interaction with architectural memory.

2024 **Minetrarchi – Minecraft-Based Architectural Learning Game** **Co-Designer · Design Concept**

Designed the concept, art, game mechanics and interactions for a Minecraft-based game concept teaching traditional Chinese timber architecture through guided tasks and free building.

EXTERNAL RESEARCH COLLABORATIONS

2025 **Eye-Tracking Experiment and Analysis Coordinator** **Tongji University & Huawei Technologies Co., Ltd.**

Provided methodological training and supervised eye-tracking experiments and data analysis for an industry collaboration on task complexity in cloud change operations.

2025–2026 **EEG Data Analysis Coordinator** **Beijing Institute of Petrochemical Technology**

Supervised and conducted EEG data analysis for the project "Cognitive and Neural Mechanisms Underlying the Detection and Dissemination of AI-Generated Misinformation."

LEADERSHIP & CULTURAL ENGAGEMENT

2020–2023 **Photojournalist; Co-Deputy Head & Head of Photography** **Jilin University Press Corps**

Led photographic reporting for campus news, cultural events and public communication.

2020–2024 **Deputy Head** **New Media Department, Physics Student Union, Jilin University**

Led college online communications and media presence; co-organised campus activities.

2020–2022 **Changchun Architectural Heritage Documentation** **Independent Researcher / Writer / Photographer**

Studied Changchun's Japanese-colonial (Manchukuo-era) architecture through public history writing and field photography; advocated preservation. Individual articles reached 10,000+ reads and 100+ saves.

TECHNICAL SKILLS

Experimental Methods	Driving Simulation • Eye Tracking • EEG/fNIRS Analysis
Research · Analysis	SPSS • MATLAB • GraphPad Prism • NVivo • Tobii Pro Lab
Programming · XR	Unity • Python • C/C++
Design · Media	Figma • Adobe Photoshop • Adobe Premiere Pro • Blender • SketchUp
Authoring	LaTeX

LANGUAGES

Chinese - Native, **English** - Academic working proficiency, **Japanese** - Intermediate, **French** - Elementary